

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 10:13:46

Training Start: 2025-05-07 09:50:36

Training End: 2025-05-07 10:00:00

Training Duration: 9.41 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.6966

MSE: 5.2262

RMSE: 2.2861

R2 Score: 0.0858

Multi-Step Forecast:

Forecast Steps: 20

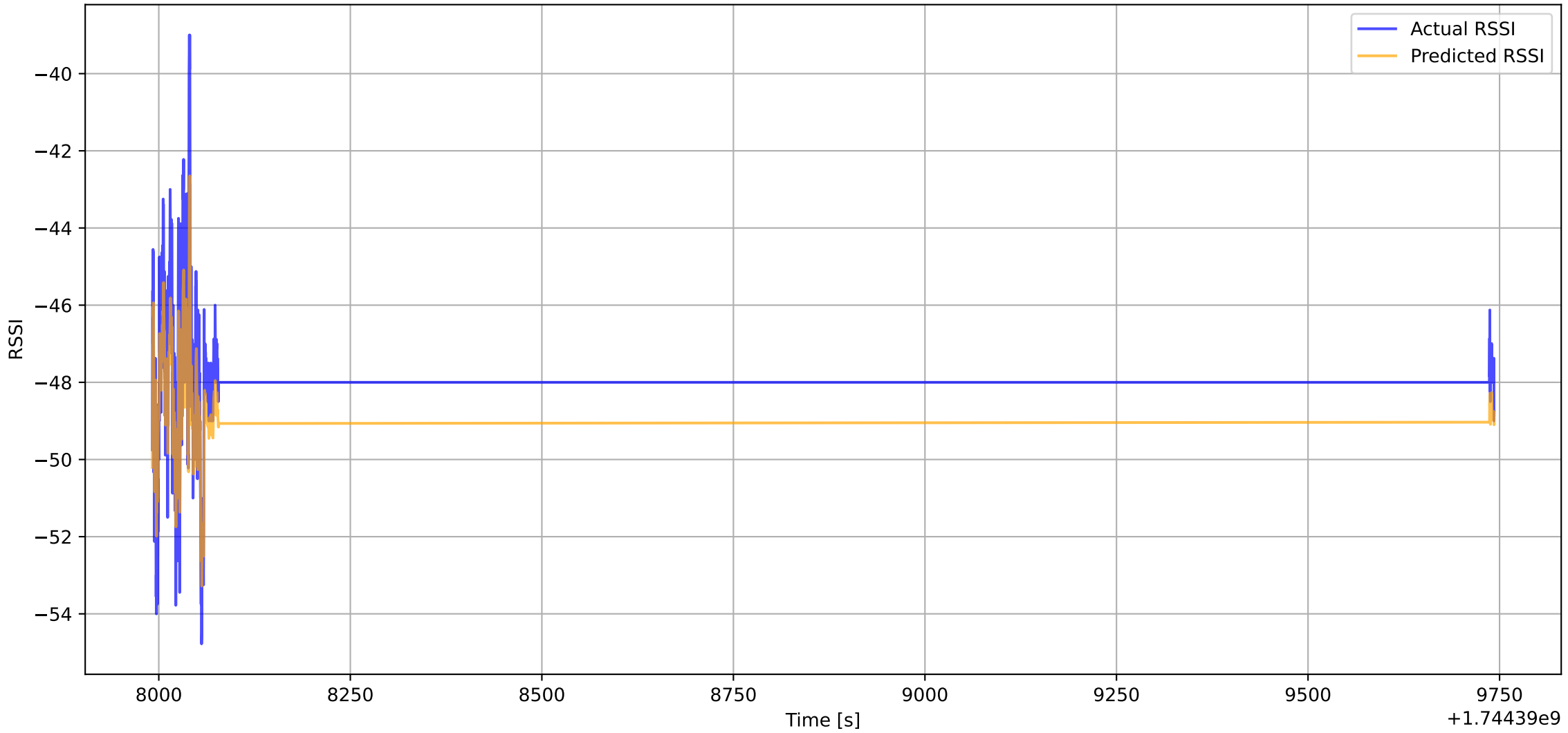
MAE: 0.0454

MSE: 0.0032

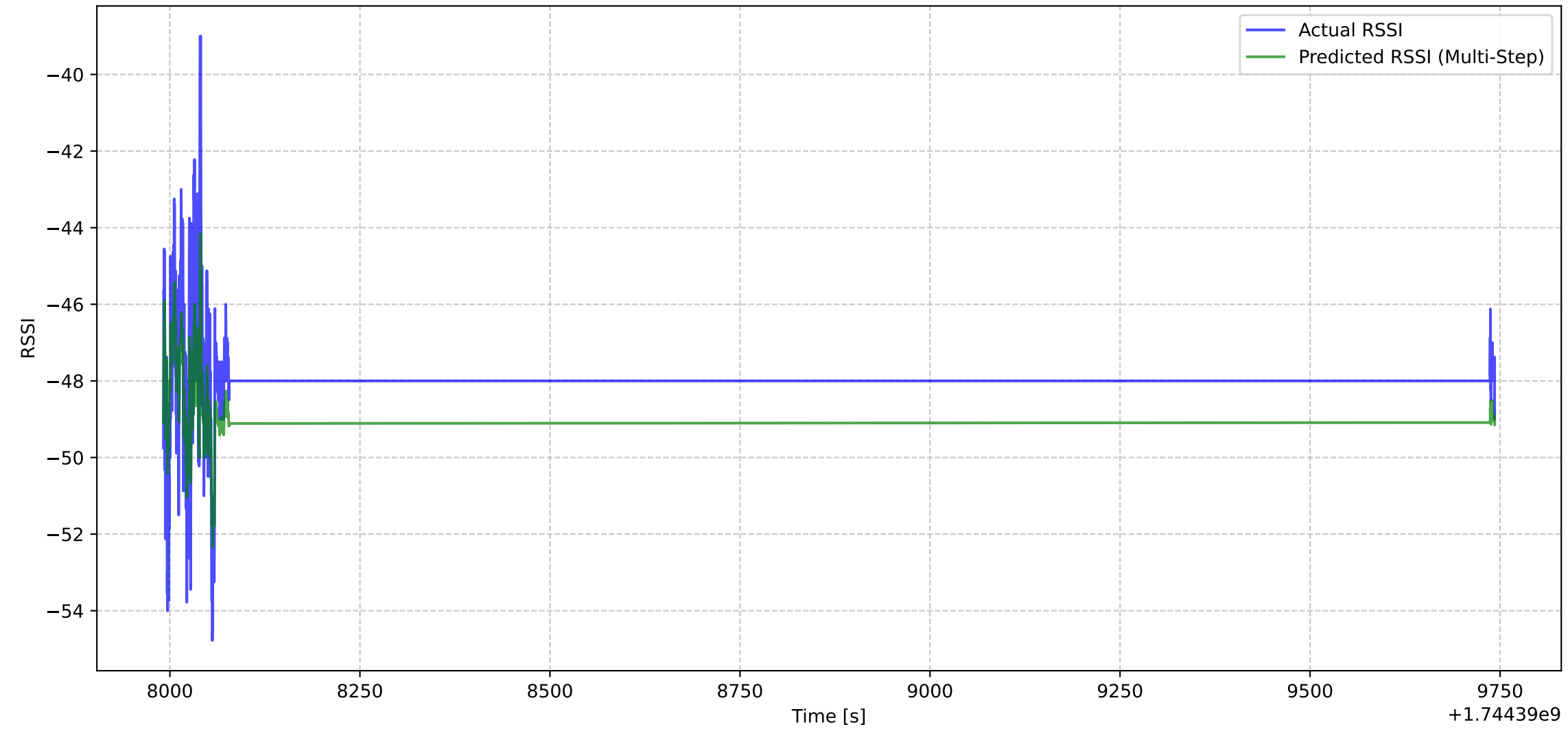
RMSE: 0.0562

R2 Score: 0.6071

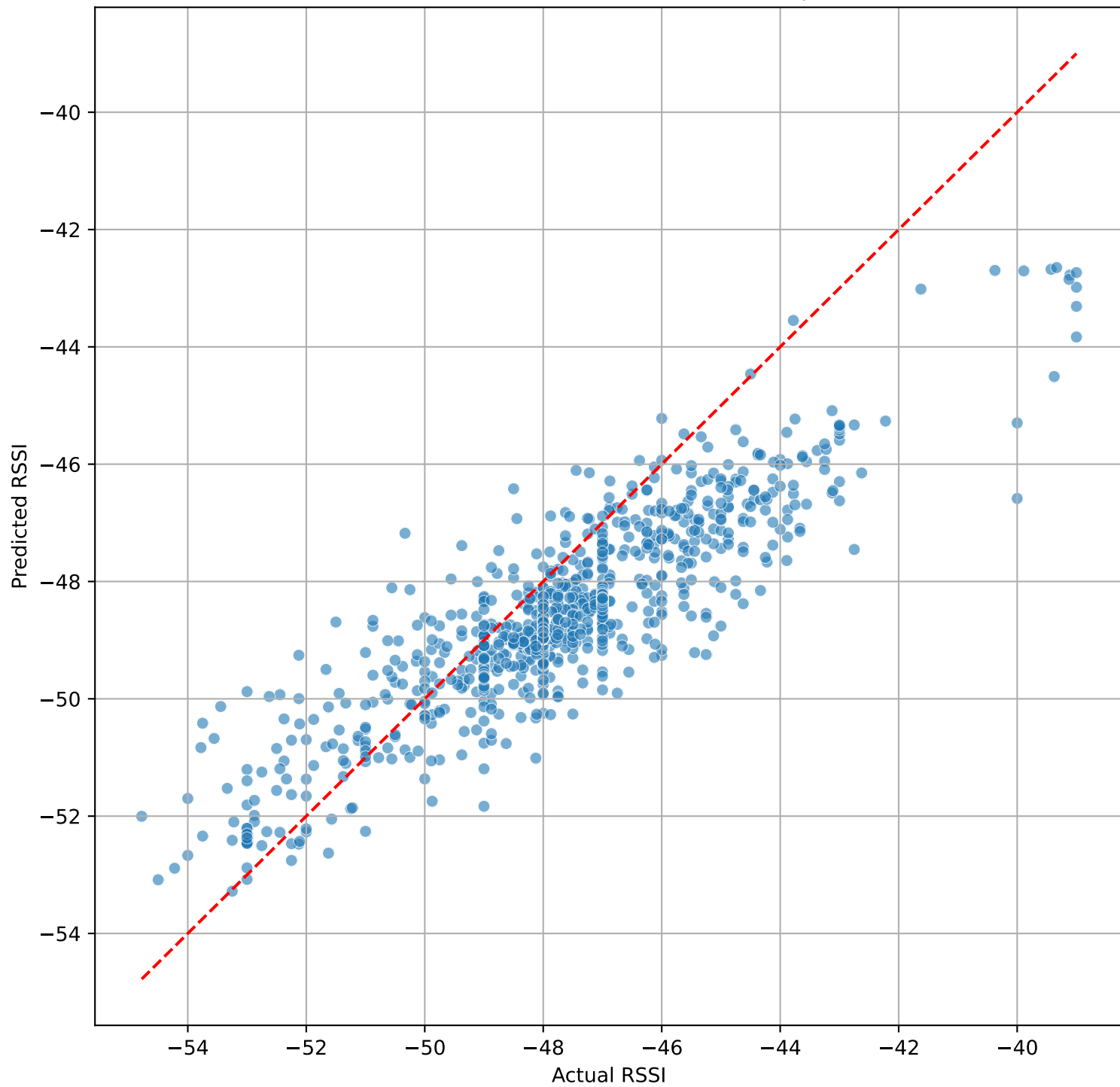
Actual vs Predicted RSSI (First Step)



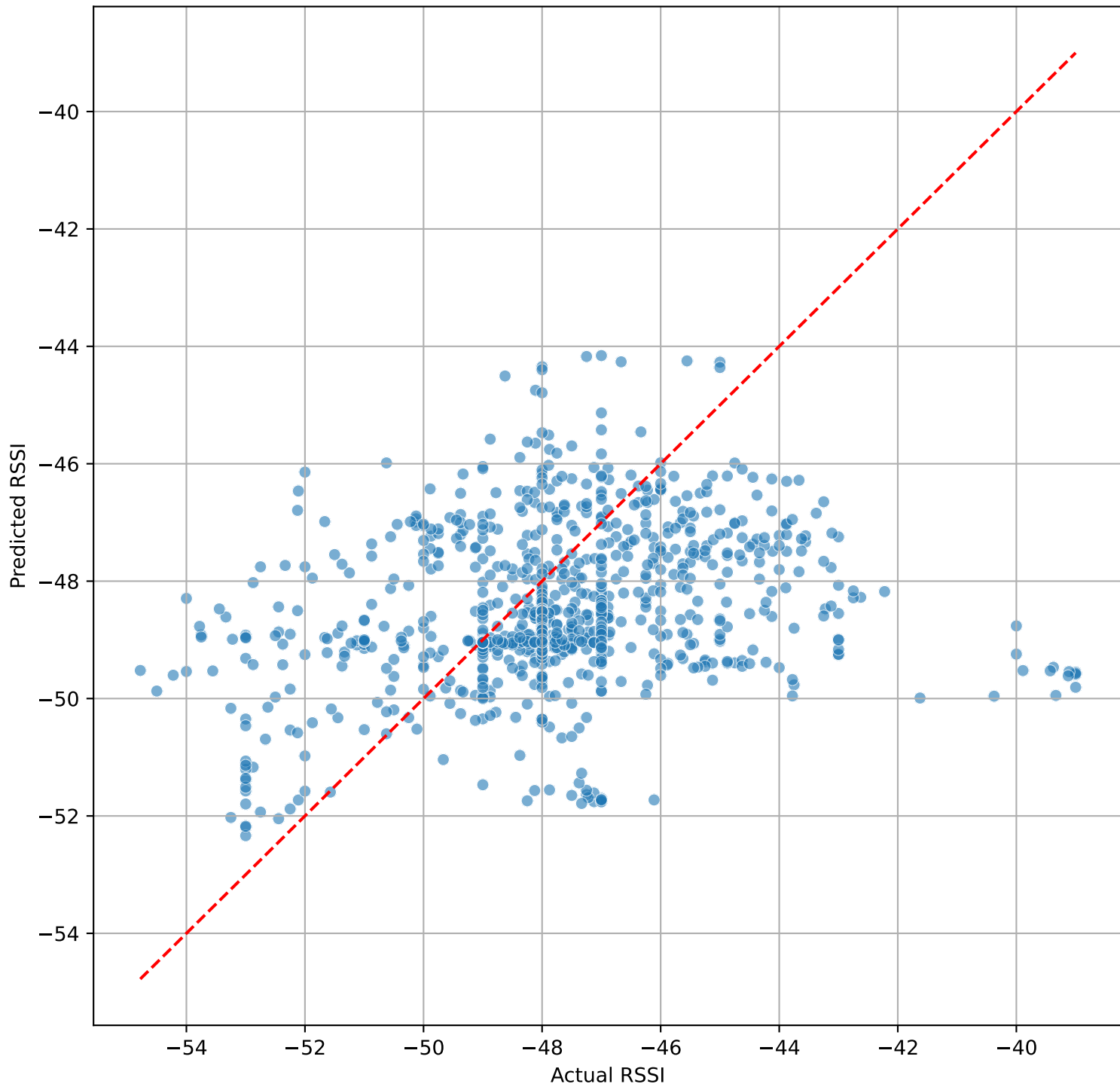
Actual vs Predicted RSSI (Multi-Step (step 20))



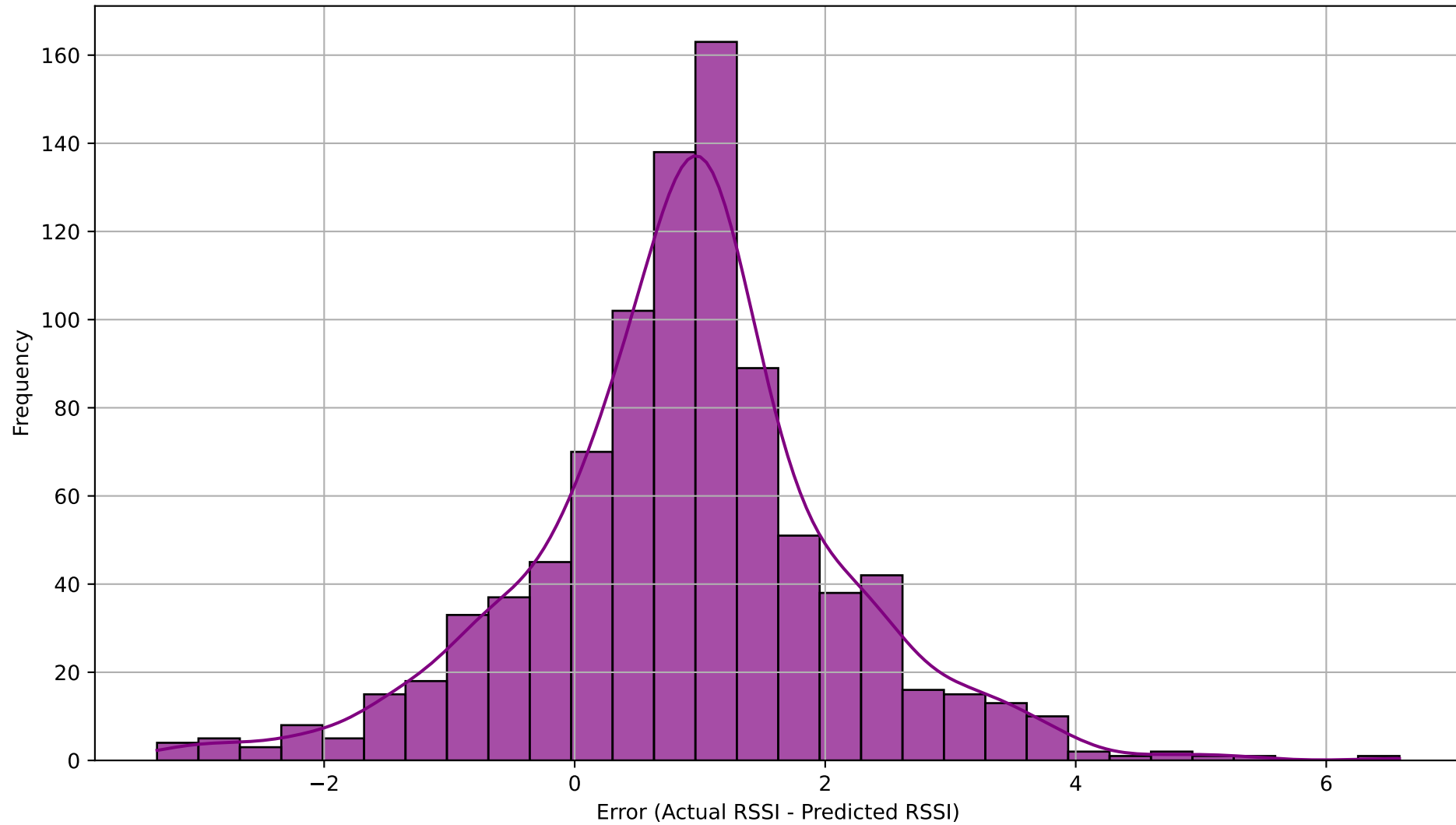
Actual vs Predicted RSSI (First Step)



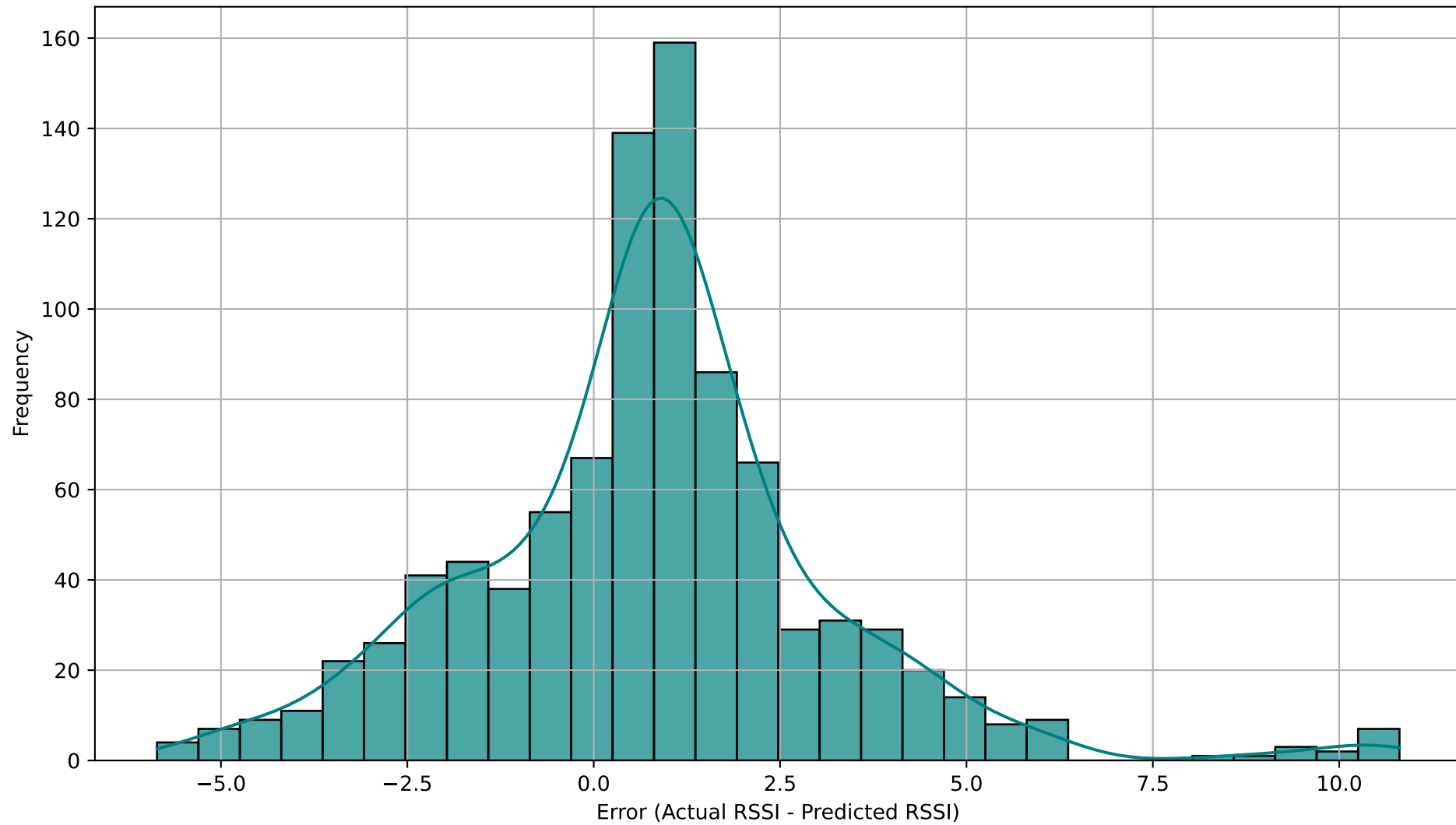
Actual vs Predicted RSSI (Last Step - step 20)



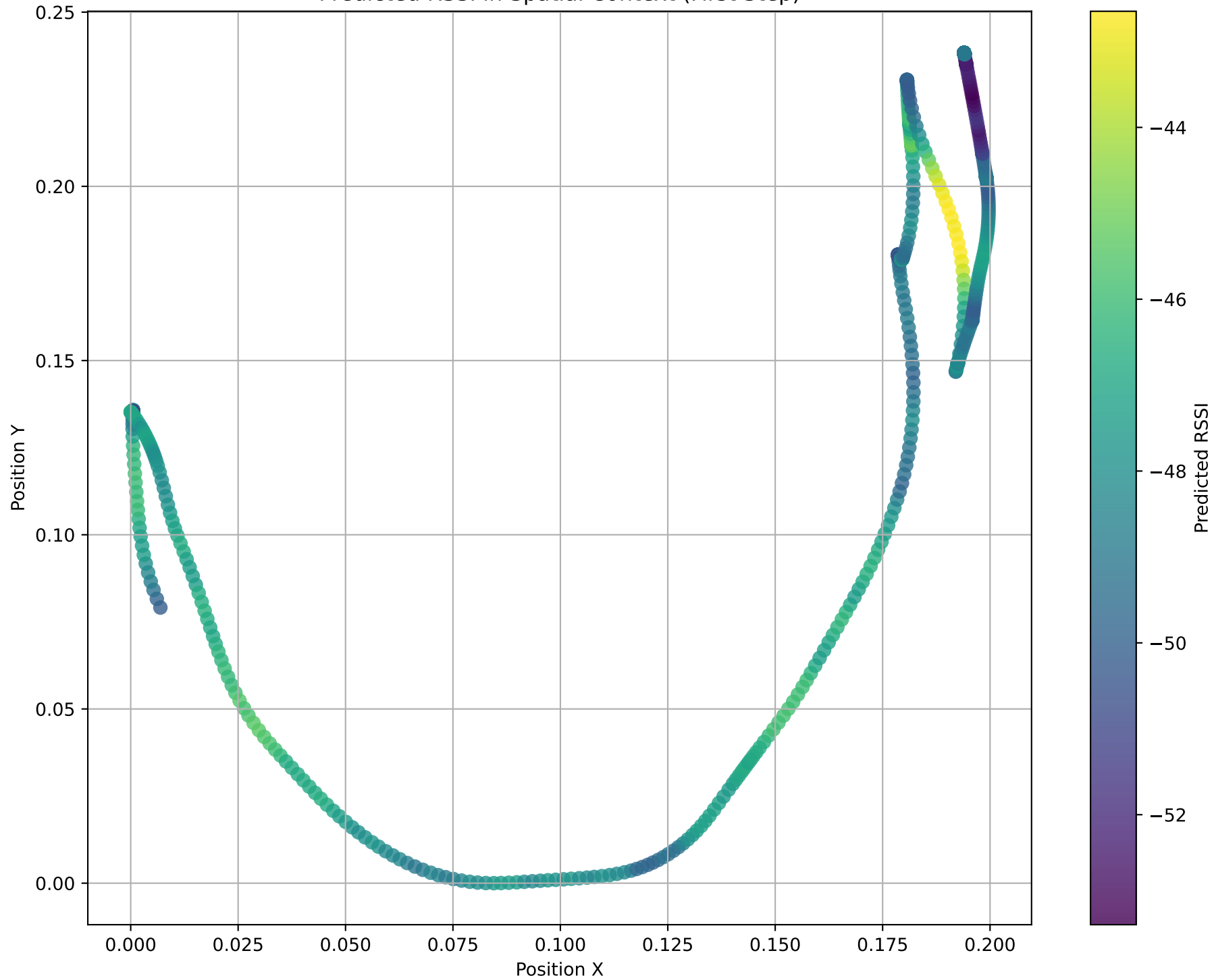
Distribution of Prediction Errors (First Step)



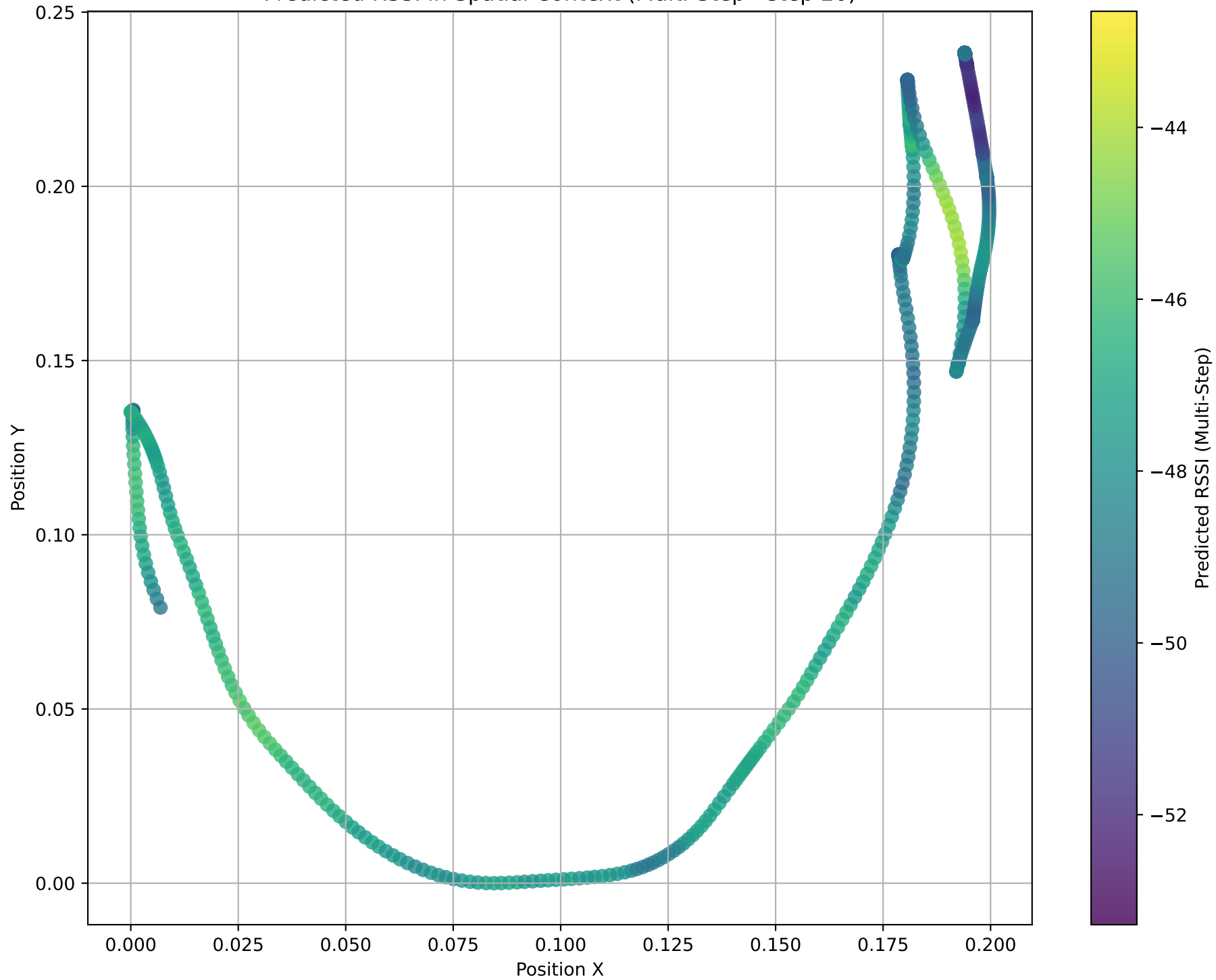
Distribution of Prediction Errors (Last Step - step 20)



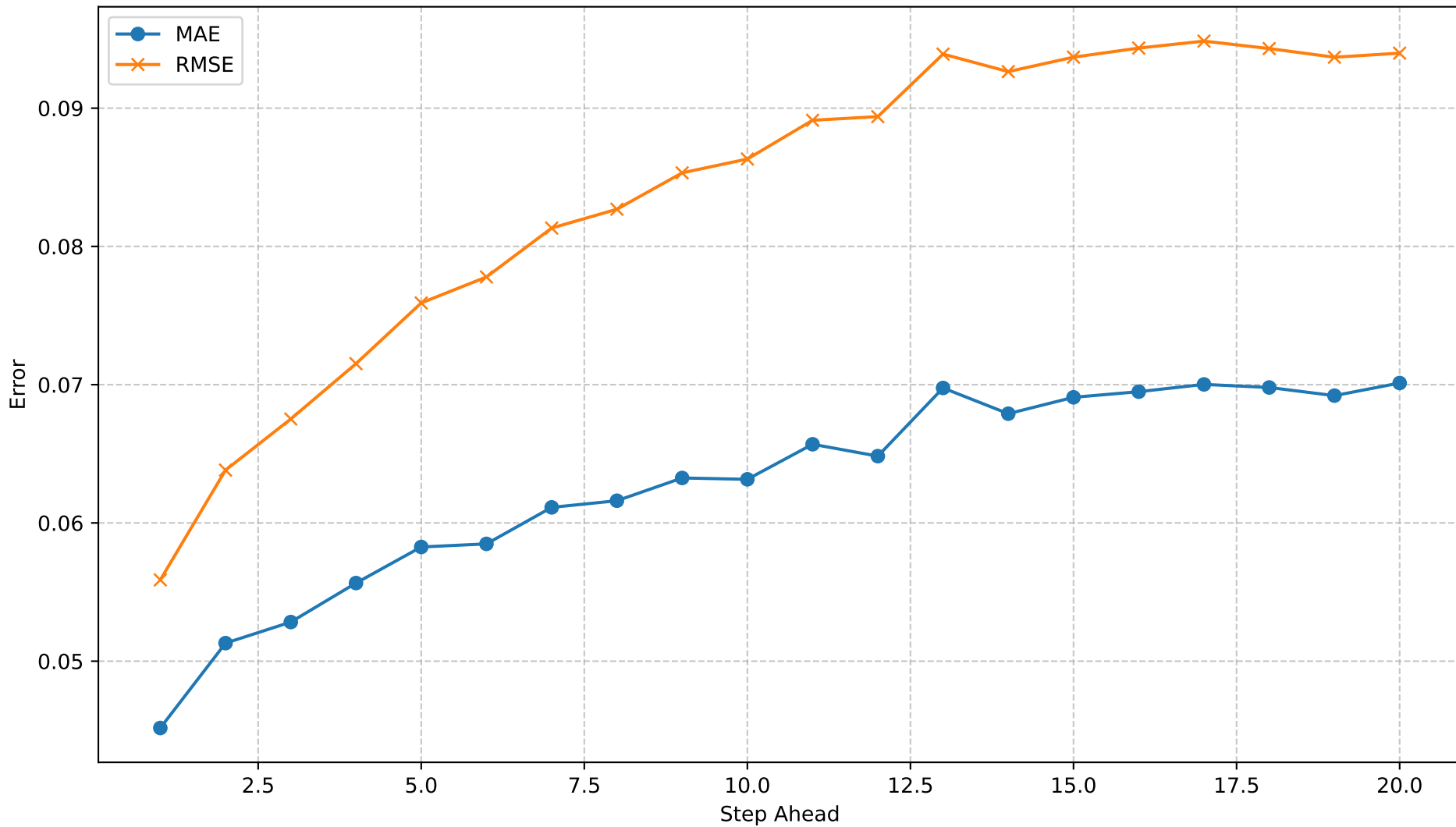
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.06
- Mediana : 0.06 / 0.09
- 20º passo : 0.07 / 0.09

Crescimento do erro: 55.2% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.07), 17 (0.07), 18 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.